Final Task: Loan Prediction Model

Prediksi Performa Kredit Peminjam (Good vs Bad Loan)

Dataset: Loan Data 2007-2014 (LendingClub) **Objective:** Membangun model Machine Learning untuk memprediksi apakah peminjam akan memiliki performa kredit yang baik (GOOD) atau buruk (BAD). **Algoritma:** Logistic Regression & Random Forest Classifier

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Alur Pengerjaan:

1. **Data Understanding** - Memahami struktur dan karakteristik data
2. **Exploratory Data Analysis (EDA)** - Visualisasi dan analisis pola data
3. **Data Preparation** - Pembersihan, encoding, dan scaling data
4. **Data Modelling** - Pelatihan model dengan Logistic Regression & Random Forest
5. **Evaluation** - Evaluasi kinerja model

1. Import Libraries

Mengimpor semua library yang dibutuhkan untuk analisis data, visualisasi, preprocessing, dan pemodelan.

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Mengubah Tema Warna Penampilan Seluruh Chart
sns.set_theme(style="whitegrid", palette='tab10', context="notebook", font_scale=1.
plt.rcParams['figure.facecolor'] = '#fafafa'
plt.rcParams['axes.facecolor'] = '#ffffff'

import warnings
warnings.filterwarnings('ignore')

from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (accuracy_score, precision_score, recall_score,
                             f1_score, roc_auc_score, roc_curve,
                             classification_report, confusion_matrix)
```

```
pd.set_option('display.max_columns', 80)
pd.set_option('display.max_rows', 100)
sns.set_style('whitegrid')
plt.rcParams['figure.figsize'] = (12, 6)
plt.rcParams['font.size'] = 12

print("Semua library berhasil diimpor!")
```

```
Semua library berhasil diimpor!
```

2. Data Understanding

2.1 Memuat Dataset

Memuat dataset loan dari file CSV dan melakukan pemeriksaan awal terhadap struktur data.

In [2]:
```
df = pd.read_csv('loan_data_2007_2014.csv', low_memory=False)
print(f"Ukuran Dataset: {df.shape[0]:,} baris x {df.shape[1]} kolom")
print(f"\n5 Baris Pertama:")
df.head()
```

```
Ukuran Dataset: 466,285 baris x 75 kolom

5 Baris Pertama:
```

Out[2]:

	Unnamed: 0	id	member_id	loan_amnt	funded_amnt	funded_amnt_inv	term	i
0	0	1077501	1296599	5000	5000	4975.0	36 months	
1	1	1077430	1314167	2500	2500	2500.0	60 months	
2	2	1077175	1313524	2400	2400	2400.0	36 months	
3	3	1076863	1277178	10000	10000	10000.0	36 months	
4	4	1075358	1311748	3000	3000	3000.0	60 months	

2.2 Informasi Dataset

Melihat tipe data dan jumlah nilai non-null untuk setiap kolom.

In [3]:

```
print("Informasi Dataset:")
print(f"Total baris: {df.shape[0]:,}")
print(f"Total kolom: {df.shape[1]}")
print(f"\nMemory usage: {df.memory_usage(deep=True).sum() / 1024**2:.2f} MB")
print("\n" + "="*60)
df.info()
```

```
Informasi Dataset:
Total baris: 466,285
Total kolom: 75

Memory usage: 779.95 MB

============================================================
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 466285 entries, 0 to 466284
Data columns (total 75 columns):
 #   Column                      Non-Null Count   Dtype
---  ------                      --------------   -----
 0   Unnamed: 0                  466285 non-null  int64
 1   id                          466285 non-null  int64
 2   member_id                   466285 non-null  int64
 3   loan_amnt                   466285 non-null  int64
 4   funded_amnt                 466285 non-null  int64
 5   funded_amnt_inv             466285 non-null  float64
 6   term                        466285 non-null  object
 7   int_rate                    466285 non-null  float64
 8   installment                 466285 non-null  float64
 9   grade                       466285 non-null  object
 10  sub_grade                   466285 non-null  object
 11  emp_title                   438697 non-null  object
 12  emp_length                  445277 non-null  object
 13  home_ownership              466285 non-null  object
 14  annual_inc                  466281 non-null  float64
 15  verification_status         466285 non-null  object
 16  issue_d                     466285 non-null  object
 17  loan_status                 466285 non-null  object
 18  pymnt_plan                  466285 non-null  object
 19  url                         466285 non-null  object
 20  desc                        125981 non-null  object
 21  purpose                     466285 non-null  object
 22  title                       466264 non-null  object
 23  zip_code                    466285 non-null  object
 24  addr_state                  466285 non-null  object
 25  dti                         466285 non-null  float64
 26  delinq_2yrs                 466256 non-null  float64
 27  earliest_cr_line            466256 non-null  object
 28  inq_last_6mths              466256 non-null  float64
 29  mths_since_last_delinq      215934 non-null  float64
 30  mths_since_last_record      62638 non-null   float64
 31  open_acc                    466256 non-null  float64
 32  pub_rec                     466256 non-null  float64
 33  revol_bal                   466285 non-null  int64
 34  revol_util                  465945 non-null  float64
 35  total_acc                   466256 non-null  float64
 36  initial_list_status         466285 non-null  object
 37  out_prncp                   466285 non-null  float64
 38  out_prncp_inv               466285 non-null  float64
 39  total_pymnt                 466285 non-null  float64
 40  total_pymnt_inv             466285 non-null  float64
 41  total_rec_prncp             466285 non-null  float64
 42  total_rec_int               466285 non-null  float64
 43  total_rec_late_fee          466285 non-null  float64
```

```
 44  recoveries                   466285 non-null  float64
 45  collection_recovery_fee      466285 non-null  float64
 46  last_pymnt_d                 465909 non-null  object
 47  last_pymnt_amnt              466285 non-null  float64
 48  next_pymnt_d                 239071 non-null  object
 49  last_credit_pull_d           466243 non-null  object
 50  collections_12_mths_ex_med   466140 non-null  float64
 51  mths_since_last_major_derog  98974 non-null   float64
 52  policy_code                  466285 non-null  int64
 53  application_type             466285 non-null  object
 54  annual_inc_joint             0 non-null       float64
 55  dti_joint                    0 non-null       float64
 56  verification_status_joint    0 non-null       float64
 57  acc_now_delinq               466256 non-null  float64
 58  tot_coll_amt                 396009 non-null  float64
 59  tot_cur_bal                  396009 non-null  float64
 60  open_acc_6m                  0 non-null       float64
 61  open_il_6m                   0 non-null       float64
 62  open_il_12m                  0 non-null       float64
 63  open_il_24m                  0 non-null       float64
 64  mths_since_rcnt_il           0 non-null       float64
 65  total_bal_il                 0 non-null       float64
 66  il_util                      0 non-null       float64
 67  open_rv_12m                  0 non-null       float64
 68  open_rv_24m                  0 non-null       float64
 69  max_bal_bc                   0 non-null       float64
 70  all_util                     0 non-null       float64
 71  total_rev_hi_lim             396009 non-null  float64
 72  inq_fi                       0 non-null       float64
 73  total_cu_tl                  0 non-null       float64
 74  inq_last_12m                 0 non-null       float64
dtypes: float64(46), int64(7), object(22)
memory usage: 266.8+ MB
```

2.3 Statistik Deskriptif

Melihat ringkasan statistik untuk kolom-kolom numerik.

In [4]:
```
df.describe().T
```

Out[4]:

	count	mean	std	min	25%
Unnamed: 0	466285.0	2.331420e+05	1.346050e+05	0.00	1.165710e+05
id	466285.0	1.307973e+07	1.089371e+07	54734.00	3.639987e+06
member_id	466285.0	1.459766e+07	1.168237e+07	70473.00	4.379705e+06
loan_amnt	466285.0	1.431728e+04	8.286509e+03	500.00	8.000000e+03
funded_amnt	466285.0	1.429180e+04	8.274371e+03	500.00	8.000000e+03
funded_amnt_inv	466285.0	1.422233e+04	8.297638e+03	0.00	8.000000e+03
int_rate	466285.0	1.382924e+01	4.357587e+00	5.42	1.099000e+01
installment	466285.0	4.320612e+02	2.434855e+02	15.67	2.566900e+02
annual_inc	466281.0	7.327738e+04	5.496357e+04	1896.00	4.500000e+04
dti	466285.0	1.721876e+01	7.851121e+00	0.00	1.136000e+01
delinq_2yrs	466256.0	2.846784e-01	7.973651e-01	0.00	0.000000e+00
inq_last_6mths	466256.0	8.047446e-01	1.091598e+00	0.00	0.000000e+00
mths_since_last_delinq	215934.0	3.410443e+01	2.177849e+01	0.00	1.600000e+01
mths_since_last_record	62638.0	7.430601e+01	3.035765e+01	0.00	5.300000e+01
open_acc	466256.0	1.118707e+01	4.987526e+00	0.00	8.000000e+00
pub_rec	466256.0	1.605642e-01	5.108626e-01	0.00	0.000000e+00
revol_bal	466285.0	1.623020e+04	2.067625e+04	0.00	6.413000e+03
revol_util	465945.0	5.617695e+01	2.373263e+01	0.00	3.920000e+01
total_acc	466256.0	2.506443e+01	1.160014e+01	1.00	1.700000e+01
out_prncp	466285.0	4.410062e+03	6.355079e+03	0.00	0.000000e+00
out_prncp_inv	466285.0	4.408452e+03	6.353198e+03	0.00	0.000000e+00
total_pymnt	466285.0	1.154069e+04	8.265627e+03	0.00	5.552125e+03
total_pymnt_inv	466285.0	1.146989e+04	8.254158e+03	0.00	5.499250e+03
total_rec_prncp	466285.0	8.866015e+03	7.031688e+03	0.00	3.708560e+03
total_rec_int	466285.0	2.588677e+03	2.483810e+03	0.00	9.572800e+02
total_rec_late_fee	466285.0	6.501292e-01	5.265730e+00	0.00	0.000000e+00
recoveries	466285.0	8.534421e+01	5.522161e+02	0.00	0.000000e+00
collection_recovery_fee	466285.0	8.961534e+00	8.549144e+01	0.00	0.000000e+00
last_pymnt_amnt	466285.0	3.123914e+03	5.554737e+03	0.00	3.126200e+02
collections_12_mths_ex_med	466140.0	9.085253e-03	1.086484e-01	0.00	0.000000e+00

	count	mean	std	min	25%
mths_since_last_major_derog	98974.0	4.285255e+01	2.166259e+01	0.00	2.600000e+01
policy_code	466285.0	1.000000e+00	0.000000e+00	1.00	1.000000e+00
annual_inc_joint	0.0	NaN	NaN	NaN	NaN
dti_joint	0.0	NaN	NaN	NaN	NaN
verification_status_joint	0.0	NaN	NaN	NaN	NaN
acc_now_delinq	466256.0	4.002093e-03	6.863680e-02	0.00	0.000000e+00
tot_coll_amt	396009.0	1.919135e+02	1.463021e+04	0.00	0.000000e+00
tot_cur_bal	396009.0	1.388017e+05	1.521147e+05	0.00	2.861800e+04
open_acc_6m	0.0	NaN	NaN	NaN	NaN
open_il_6m	0.0	NaN	NaN	NaN	NaN
open_il_12m	0.0	NaN	NaN	NaN	NaN
open_il_24m	0.0	NaN	NaN	NaN	NaN
mths_since_rcnt_il	0.0	NaN	NaN	NaN	NaN
total_bal_il	0.0	NaN	NaN	NaN	NaN
il_util	0.0	NaN	NaN	NaN	NaN
open_rv_12m	0.0	NaN	NaN	NaN	NaN
open_rv_24m	0.0	NaN	NaN	NaN	NaN
max_bal_bc	0.0	NaN	NaN	NaN	NaN
all_util	0.0	NaN	NaN	NaN	NaN
total_rev_hi_lim	396009.0	3.037909e+04	3.724713e+04	0.00	1.350000e+04
inq_fi	0.0	NaN	NaN	NaN	NaN
total_cu_tl	0.0	NaN	NaN	NaN	NaN
inq_last_12m	0.0	NaN	NaN	NaN	NaN

2.4 Pemeriksaan Missing Values

Mengidentifikasi kolom-kolom yang memiliki nilai kosong (missing values).

In [5]:
```
missing = df.isnull().sum()
missing_pct = (missing / len(df) * 100).round(2)
missing_df = pd.DataFrame({'Missing Count': missing, 'Missing %': missing_pct})
missing_df = missing_df[missing_df['Missing Count'] > 0].sort_values('Missing %', a
print(f"Jumlah kolom dengan missing values: {len(missing_df)} dari {df.shape[1]} ko
print(missing_df.to_string())
```

```
Jumlah kolom dengan missing values: 40 dari 75 kolom

                            Missing Count  Missing %
dti_joint                          466285     100.00
annual_inc_joint                   466285     100.00
total_cu_tl                        466285     100.00
inq_fi                             466285     100.00
all_util                           466285     100.00
max_bal_bc                         466285     100.00
open_rv_24m                        466285     100.00
open_rv_12m                        466285     100.00
il_util                            466285     100.00
total_bal_il                       466285     100.00
mths_since_rcnt_il                 466285     100.00
open_il_24m                        466285     100.00
open_il_12m                        466285     100.00
open_il_6m                         466285     100.00
open_acc_6m                        466285     100.00
verification_status_joint          466285     100.00
inq_last_12m                       466285     100.00
mths_since_last_record             403647      86.57
mths_since_last_major_derog        367311      78.77
desc                               340304      72.98
mths_since_last_delinq             250351      53.69
next_pymnt_d                       227214      48.73
total_rev_hi_lim                    70276      15.07
tot_coll_amt                        70276      15.07
tot_cur_bal                         70276      15.07
emp_title                           27588       5.92
emp_length                          21008       4.51
last_pymnt_d                          376       0.08
revol_util                            340       0.07
collections_12_mths_ex_med            145       0.03
open_acc                               29       0.01
total_acc                              29       0.01
inq_last_6mths                         29       0.01
earliest_cr_line                       29       0.01
delinq_2yrs                            29       0.01
acc_now_delinq                         29       0.01
last_credit_pull_d                     42       0.01
pub_rec                                29       0.01
title                                  21       0.00
annual_inc                              4       0.00
```

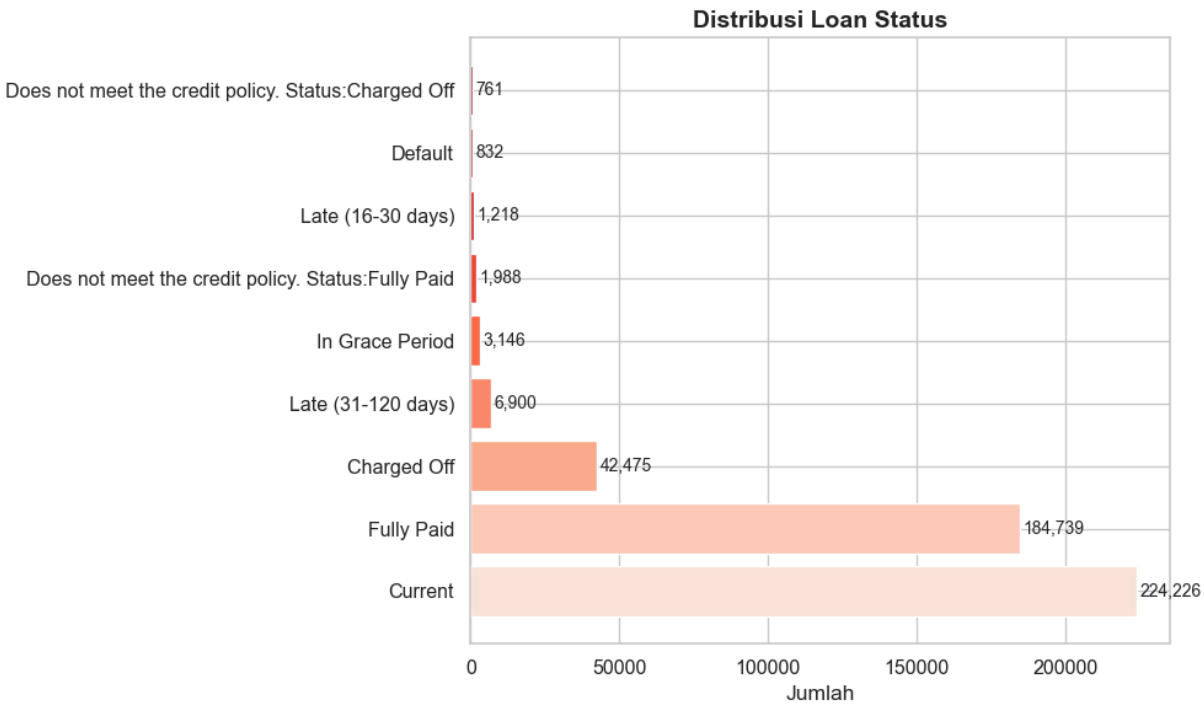
2.5 Distribusi Target Variable (loan_status)

Melihat distribusi nilai pada kolom `loan_status` yang akan digunakan sebagai dasar pembuatan label target.

In [6]:
```
print("Distribusi Loan Status:\n")
status_counts = df['loan_status'].value_counts()
for status, count in status_counts.items():
    pct = count / len(df) * 100
    print(f"  {status:55s} -> {count:>8,} ({pct:.2f}%)")
```

```
fig, ax = plt.subplots(figsize=(10, 6))
colors = sns.color_palette("Reds", len(status_counts))
bars = ax.barh(status_counts.index, status_counts.values, color=colors)
ax.set_xlabel('Jumlah')
ax.set_title('Distribusi Loan Status', fontsize=14, fontweight='bold')
for bar, val in zip(bars, status_counts.values):
    ax.text(bar.get_width() + 1000, bar.get_y() + bar.get_height()/2,
            f'{val:,}', va='center', fontsize=10)
plt.tight_layout()
plt.show()
```

```
Distribusi Loan Status:

  Current                                                 ->  224,226 (48.09%)
  Fully Paid                                              ->  184,739 (39.62%)
  Charged Off                                             ->   42,475 (9.11%)
  Late (31-120 days)                                      ->    6,900 (1.48%)
  In Grace Period                                         ->    3,146 (0.67%)
  Does not meet the credit policy. Status:Fully Paid      ->    1,988 (0.43%)
  Late (16-30 days)                                       ->    1,218 (0.26%)
  Default                                                 ->      832 (0.18%)
  Does not meet the credit policy. Status:Charged Off     ->      761 (0.16%)
```



3. Pembuatan Label Target (Good vs Bad Loan)

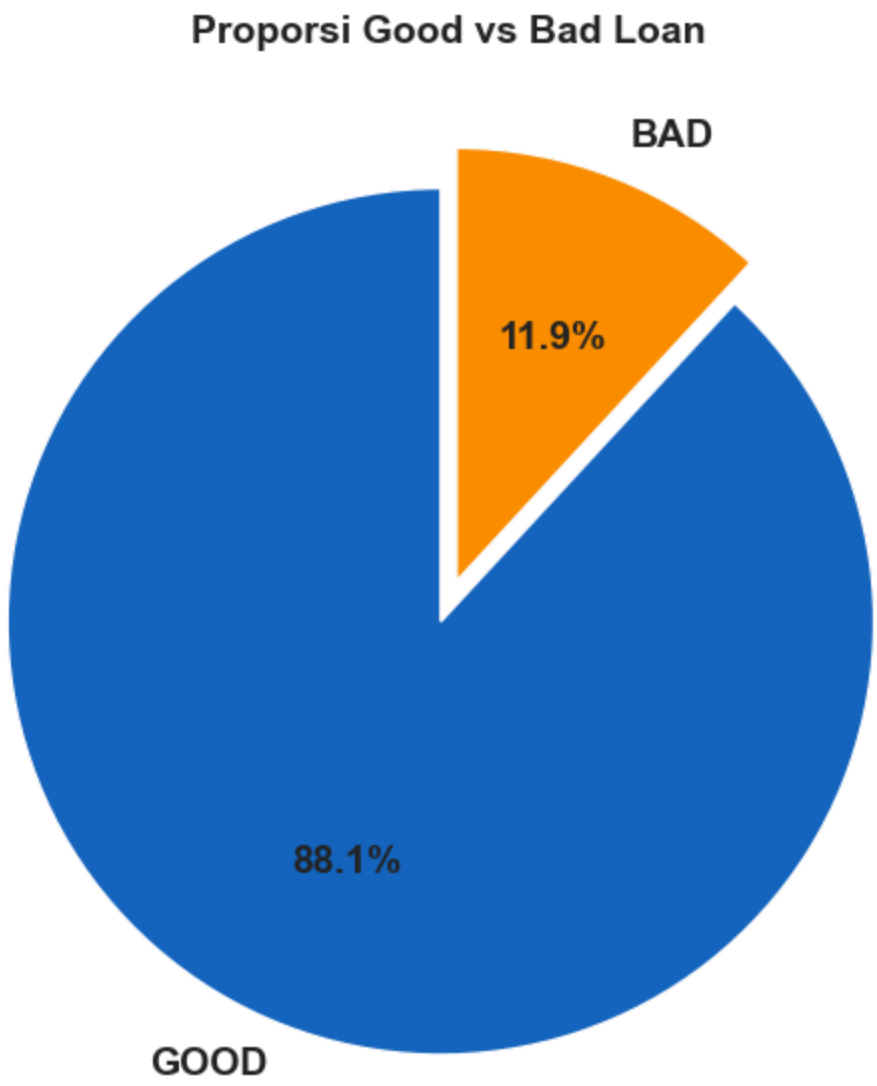
Definisi Label:

- **GOOD (1):** Peminjam dengan performa kredit baik -> `Fully Paid`, `Current`, `Does not meet the credit policy. Status:Fully Paid`
- **BAD (0):** Peminjam dengan performa kredit buruk -> `Charged Off`, `Default`, `Late (31-120 days)`, `Late (16-30 days)`, `In Grace Period`, `Does not meet the credit policy. Status:Charged Off`

Penentuan ini berdasarkan sejarah pembayaran kredit peminjam.

In [7]:
```
good_status = ['Fully Paid', 'Current', 'Does not meet the credit policy. Status:Fu
bad_status = ['Charged Off', 'Default', 'Late (31-120 days)', 'Late (16-30 days)',
              'In Grace Period', 'Does not meet the credit policy. Status:Charged O

df['loan_label'] = df['loan_status'].apply(lambda x: 1 if x in good_status else 0)

print("Distribusi Label Target:\n")
label_counts = df['loan_label'].value_counts()
print(f"  GOOD (1): {label_counts[1]:>8,} ({label_counts[1]/len(df)*100:.2f}%)")
print(f"  BAD  (0): {label_counts[0]:>8,} ({label_counts[0]/len(df)*100:.2f}%)")

fig, ax = plt.subplots(figsize=(6, 6))
colors_pie = ['#1565C0', '#FB8C00']
ax.pie(label_counts.values, labels=['GOOD', 'BAD'], autopct='%1.1f%%',
       colors=colors_pie, startangle=90, explode=(0.05, 0.05),
       textprops={'fontsize': 14, 'fontweight': 'bold'})
ax.set_title('Proporsi Good vs Bad Loan', fontsize=14, fontweight='bold')
plt.tight_layout()
plt.show()
```

```
Distribusi Label Target:

  GOOD (1):  410,953 (88.13%)
  BAD  (0):   55,332 (11.87%)
```



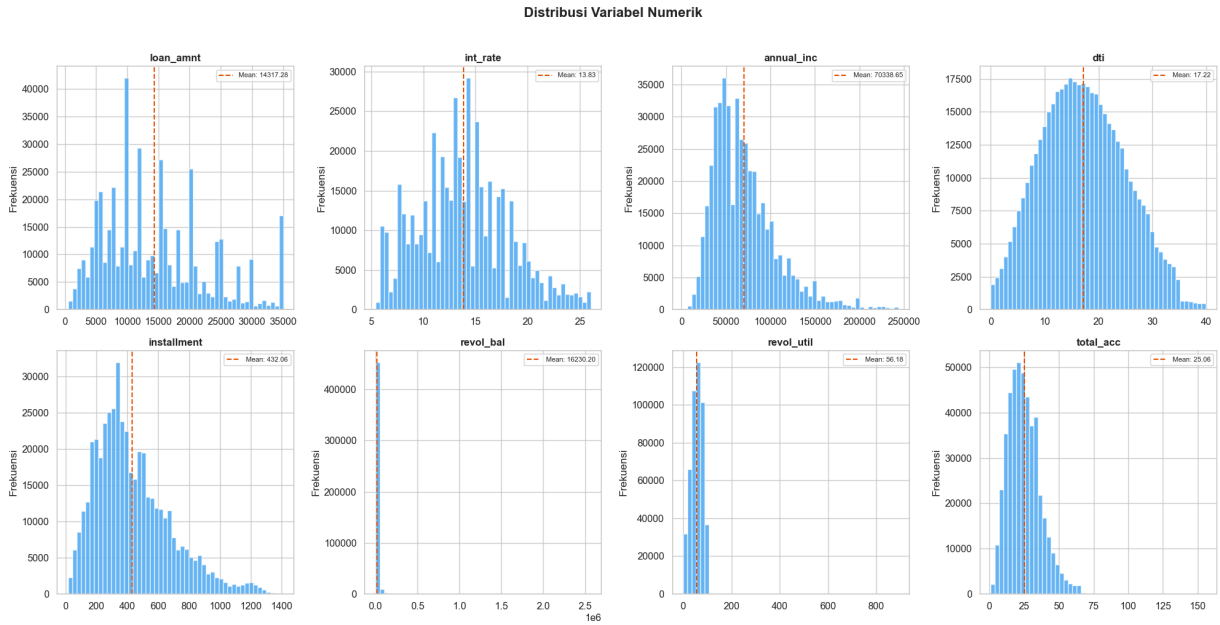
4. Exploratory Data Analysis (EDA)

4.1 Analisis Univariat - Distribusi Variabel Numerik

Menganalisis distribusi beberapa variabel numerik penting dalam dataset.

In [8]:
```
num_cols = ['loan_amnt', 'int_rate', 'annual_inc', 'dti', 'installment', 'revol_bal
available_cols = [c for c in num_cols if c in df.columns]

fig, axes = plt.subplots(2, 4, figsize=(20, 10))
axes = axes.flatten()
for i, col in enumerate(available_cols):
    data = df[col].dropna()
    if col == 'annual_inc':
        data = data[data < data.quantile(0.99)]
    axes[i].hist(data, bins=50, color='#42A5F5', edgecolor='white', alpha=0.8)
    axes[i].set_title(col, fontsize=12, fontweight='bold')
    axes[i].set_ylabel('Frekuensi')
```

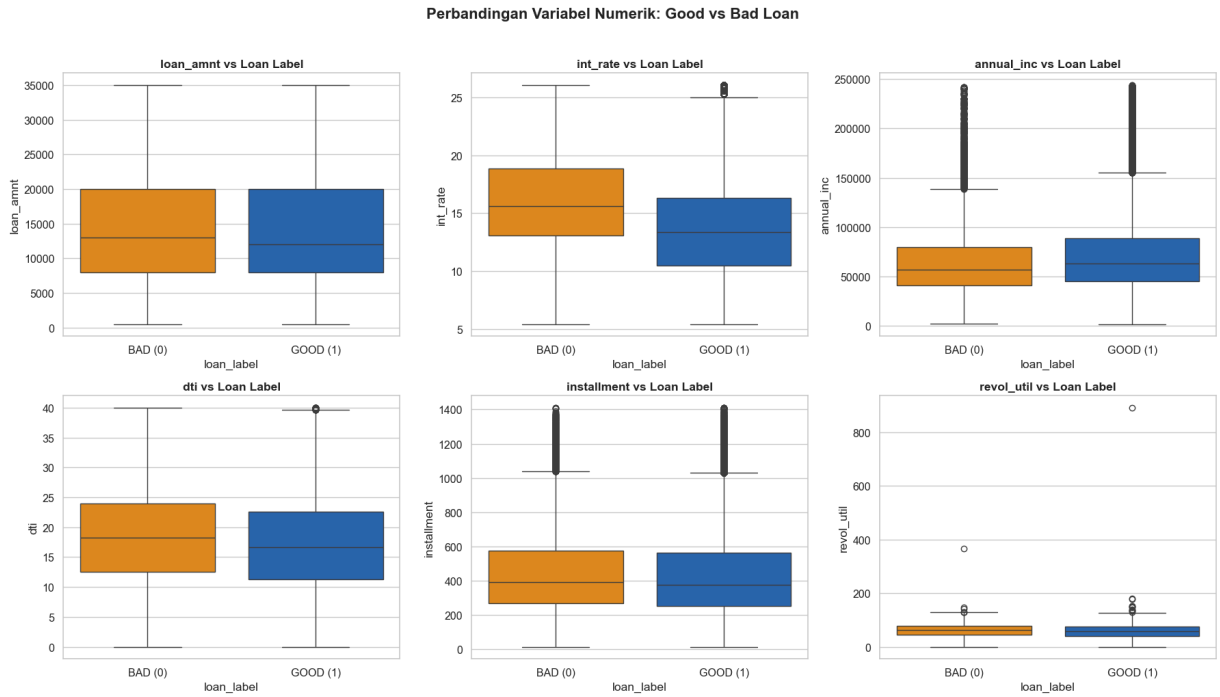
```
    axes[i].axvline(data.mean(), color='#E65100', linestyle='--', label=f'Mean: {da
    axes[i].legend(fontsize=8)
plt.suptitle('Distribusi Variabel Numerik', fontsize=16, fontweight='bold', y=1.02)
plt.tight_layout()
plt.show()
```



4.2 Analisis Bivariat - Hubungan Variabel dengan Label Target

Melihat hubungan antara beberapa variabel penting dengan status pinjaman (Good/Bad).

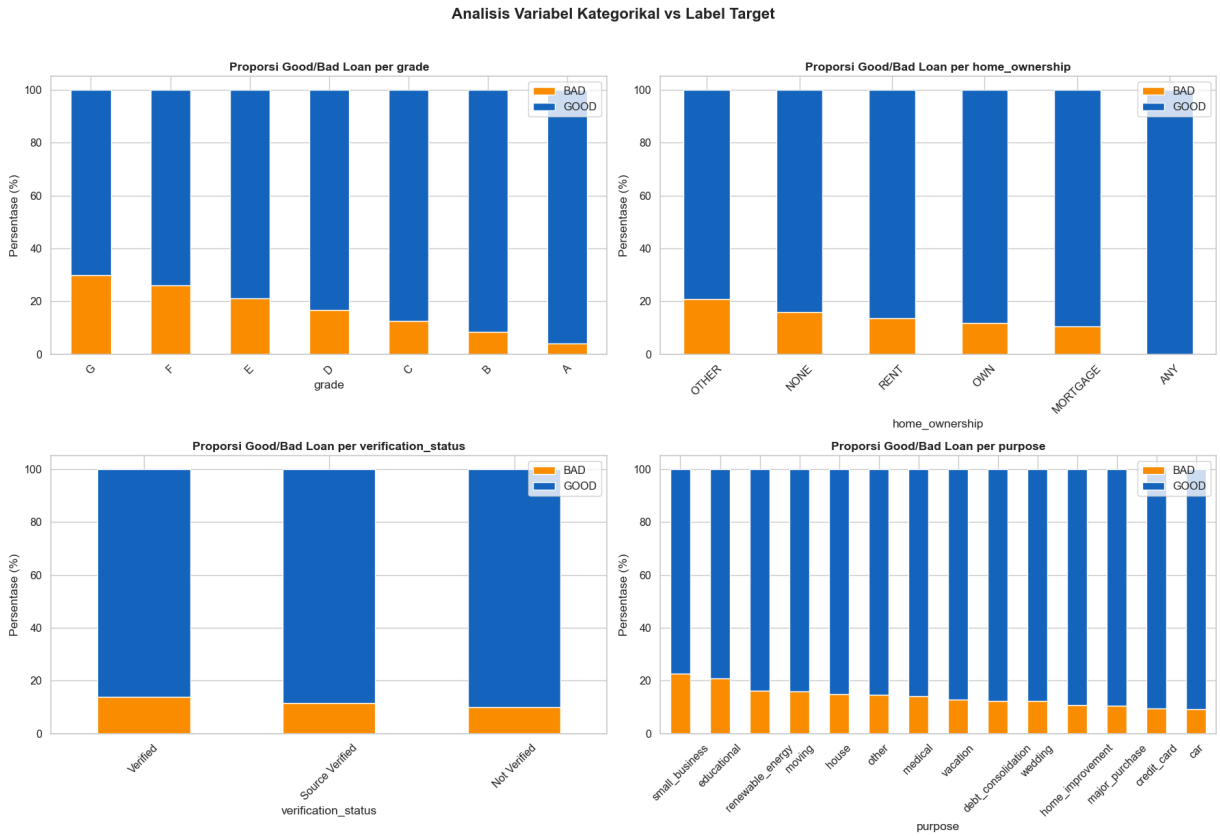
In [9]:
```
fig, axes = plt.subplots(2, 3, figsize=(18, 10))
plot_cols = ['loan_amnt', 'int_rate', 'annual_inc', 'dti', 'installment', 'revol_ut
available_plot = [c for c in plot_cols if c in df.columns]

for i, col in enumerate(available_plot):
    row, col_idx = i // 3, i % 3
    data = df[[col, 'loan_label']].dropna()
    if col == 'annual_inc':
        data = data[data[col] < data[col].quantile(0.99)]
    sns.boxplot(x='loan_label', y=col, data=data, ax=axes[row][col_idx],
                palette=['#FB8C00', '#1565C0'])
    axes[row][col_idx].set_title(f'{col} vs Loan Label', fontweight='bold')
    axes[row][col_idx].set_xticklabels(['BAD (0)', 'GOOD (1)'])
plt.suptitle('Perbandingan Variabel Numerik: Good vs Bad Loan', fontsize=16, fontwe
plt.tight_layout()
plt.show()
```



4.3 Analisis Variabel Kategorikal

Melihat distribusi variabel kategorikal terhadap label target.

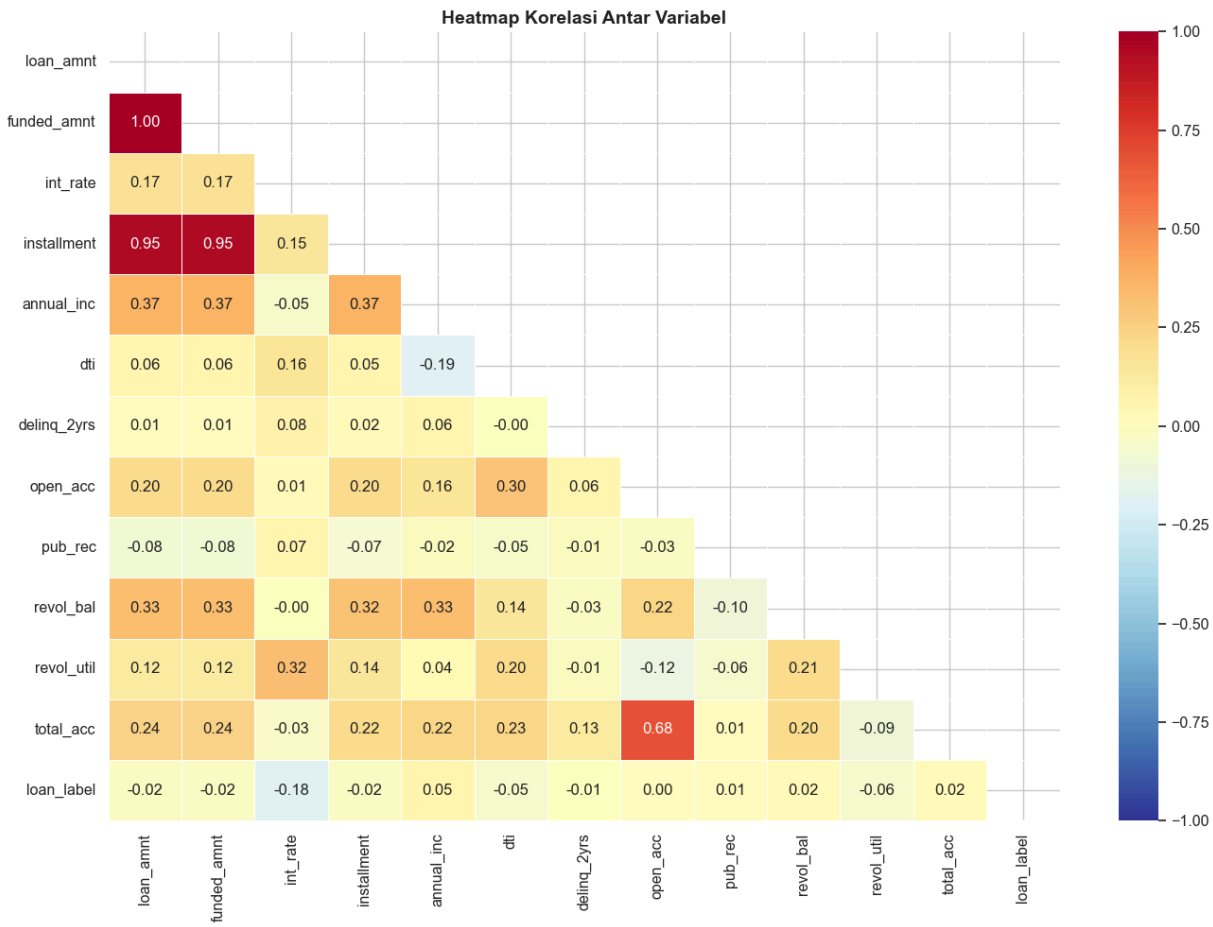
In [10]:
```
cat_cols = ['grade', 'home_ownership', 'verification_status', 'purpose']

fig, axes = plt.subplots(2, 2, figsize=(18, 12))
axes = axes.flatten()

for i, col in enumerate(cat_cols):
    ct = pd.crosstab(df[col], df['loan_label'], normalize='index') * 100
    ct = ct.sort_values(0, ascending=False)
    ct.plot(kind='bar', stacked=True, ax=axes[i], color=['#FB8C00', '#1565C0'], edg
    axes[i].set_title(f'Proporsi Good/Bad Loan per {col}', fontweight='bold')
    axes[i].set_ylabel('Persentase (%)')
    axes[i].legend(['BAD', 'GOOD'], loc='upper right')
    axes[i].tick_params(axis='x', rotation=45)

plt.suptitle('Analisis Variabel Kategorikal vs Label Target', fontsize=16, fontweig
plt.tight_layout()
plt.show()
```



4.4 Analisis Korelasi

Melihat korelasi antar variabel numerik menggunakan heatmap.

In [11]:
```
corr_cols = ['loan_amnt', 'funded_amnt', 'int_rate', 'installment', 'annual_inc',
             'dti', 'delinq_2yrs', 'open_acc', 'pub_rec', 'revol_bal',
             'revol_util', 'total_acc', 'loan_label']
available_corr = [c for c in corr_cols if c in df.columns]

corr_matrix = df[available_corr].corr()

fig, ax = plt.subplots(figsize=(14, 10))
mask = np.triu(np.ones_like(corr_matrix, dtype=bool))
sns.heatmap(corr_matrix, mask=mask, annot=True, fmt='.2f', cmap='RdYlBu_r',
            center=0, linewidths=0.5, ax=ax, vmin=-1, vmax=1)
ax.set_title('Heatmap Korelasi Antar Variabel', fontsize=14, fontweight='bold')
plt.tight_layout()
plt.show()
```



5. Data Preparation

5.1 Seleksi Fitur

Memilih fitur-fitur yang relevan untuk pemodelan dan menghapus kolom yang tidak diperlukan.

Kolom yang dihapus:

- Kolom identitas: `id`, `member_id`, `url`, `desc`, `title`, `emp_title`, `zip_code`
- Kolom status yang sudah digunakan sebagai label: `loan_status`
- Kolom post-loan (data yang hanya tersedia setelah pinjaman diberikan): `out_prncp`, `total_pymnt`, `recoveries`, dll.
- Kolom joint application (terlalu banyak missing): `annual_inc_joint`, `dti_joint`, dll.
- Kolom dengan data sangat sparse: `open_acc_6m`, `open_il_6m`, dll.

In [12]:
```
drop_cols = ['Unnamed: 0', 'id', 'member_id', 'url', 'desc', 'title', 'emp_title',
             'zip_code', 'loan_status', 'pymnt_plan', 'policy_code', 'application_t
             'annual_inc_joint', 'dti_joint', 'verification_status_joint',
             'open_acc_6m', 'open_il_6m', 'open_il_12m', 'open_il_24m',
             'mths_since_rcnt_il', 'total_bal_il', 'il_util', 'open_rv_12m',
             'open_rv_24m', 'max_bal_bc', 'all_util', 'inq_fi', 'total_cu_tl', 'inq
             'out_prncp', 'out_prncp_inv', 'total_pymnt', 'total_pymnt_inv',
```

```
                'total_rec_prncp', 'total_rec_int', 'total_rec_late_fee',
                'recoveries', 'collection_recovery_fee', 'last_pymnt_d',
                'last_pymnt_amnt', 'next_pymnt_d', 'last_credit_pull_d']

existing_drop = [c for c in drop_cols if c in df.columns]
df_clean = df.drop(columns=existing_drop)
print(f"Kolom sebelum seleksi: {df.shape[1]}")
print(f"Kolom setelah seleksi: {df_clean.shape[1]}")
print(f"\nKolom yang digunakan:\n{list(df_clean.columns)}")
```

```
Kolom sebelum seleksi: 76
Kolom setelah seleksi: 34

Kolom yang digunakan:
['loan_amnt', 'funded_amnt', 'funded_amnt_inv', 'term', 'int_rate', 'installment',
'grade', 'sub_grade', 'emp_length', 'home_ownership', 'annual_inc', 'verification_st
atus', 'issue_d', 'purpose', 'addr_state', 'dti', 'delinq_2yrs', 'earliest_cr_line',
'inq_last_6mths', 'mths_since_last_delinq', 'mths_since_last_record', 'open_acc', 'p
ub_rec', 'revol_bal', 'revol_util', 'total_acc', 'initial_list_status', 'collections
_12_mths_ex_med', 'mths_since_last_major_derog', 'acc_now_delinq', 'tot_coll_amt',
'tot_cur_bal', 'total_rev_hi_lim', 'loan_label']
```

5.2 Penanganan Missing Values

Mengisi atau menghapus missing values dengan teknik yang sesuai.

Strategi:

- Kolom dengan >50% missing values: **dihapus**
- Kolom numerik: diisi dengan **median** (lebih robust terhadap outlier)
- Kolom kategorikal: diisi dengan **modus** (nilai yang paling sering muncul)

In [13]:
```
missing_clean = df_clean.isnull().sum()
missing_clean = missing_clean[missing_clean > 0].sort_values(ascending=False)
print(f"Kolom dengan missing values: {len(missing_clean)}\n")
print(missing_clean)
```

```
Kolom dengan missing values: 17

mths_since_last_record         403647
mths_since_last_major_derog    367311
mths_since_last_delinq         250351
total_rev_hi_lim                70276
tot_cur_bal                     70276
tot_coll_amt                    70276
emp_length                      21008
revol_util                        340
collections_12_mths_ex_med        145
open_acc                           29
total_acc                          29
inq_last_6mths                     29
acc_now_delinq                     29
earliest_cr_line                   29
delinq_2yrs                        29
pub_rec                            29
annual_inc                          4
dtype: int64
```

In [14]:

```
# Hapus kolom dengan >50% missing
high_missing = [col for col in df_clean.columns if df_clean[col].isnull().sum() / l
print(f"Kolom dengan >50% missing (dihapus): {high_missing}")
df_clean = df_clean.drop(columns=high_missing)

# Isi missing values
for col in df_clean.columns:
    if df_clean[col].isnull().sum() > 0:
        if df_clean[col].dtype in ['float64', 'int64']:
            df_clean[col] = df_clean[col].fillna(df_clean[col].median())
        else:
            df_clean[col] = df_clean[col].fillna(df_clean[col].mode()[0])

print(f"\nTotal missing values setelah penanganan: {df_clean.isnull().sum().sum()}"
print(f"Shape data: {df_clean.shape}")
```

```
Kolom dengan >50% missing (dihapus): ['mths_since_last_delinq', 'mths_since_last_rec
ord', 'mths_since_last_major_derog']

Total missing values setelah penanganan: 0
Shape data: (466285, 31)
```

5.3 Feature Engineering & Encoding

Melakukan encoding pada variabel kategorikal menggunakan Label Encoding agar dapat diproses oleh model machine learning.

In [15]:

```
cat_columns = df_clean.select_dtypes(include=['object']).columns.tolist()
num_columns = df_clean.select_dtypes(include=['int64', 'float64']).columns.tolist()
if 'loan_label' in num_columns:
    num_columns.remove('loan_label')

print(f"Kolom Kategorikal ({len(cat_columns)}): {cat_columns}")
print(f"Kolom Numerik: {len(num_columns)} kolom")
```

```
Kolom Kategorikal (11): ['term', 'grade', 'sub_grade', 'emp_length', 'home_ownershi
p', 'verification_status', 'issue_d', 'purpose', 'addr_state', 'earliest_cr_line',
'initial_list_status']
Kolom Numerik: 19 kolom
```

In [16]:
```
le_dict = {}
for col in cat_columns:
    le = LabelEncoder()
    df_clean[col] = le.fit_transform(df_clean[col].astype(str))
    le_dict[col] = le
    print(f"  {col}: {len(le.classes_)} kategori di-encode")

print(f"\nShape data setelah encoding: {df_clean.shape}")
```

```
  term: 2 kategori di-encode
  grade: 7 kategori di-encode
  sub_grade: 35 kategori di-encode
  emp_length: 11 kategori di-encode
  home_ownership: 6 kategori di-encode
  verification_status: 3 kategori di-encode
  issue_d: 91 kategori di-encode
  purpose: 14 kategori di-encode
  addr_state: 50 kategori di-encode
  earliest_cr_line: 664 kategori di-encode
  initial_list_status: 2 kategori di-encode

Shape data setelah encoding: (466285, 31)
```

5.4 Penanganan Outlier

Mendeteksi dan menangani outlier pada variabel numerik menggunakan metode IQR (Interquartile Range). Outlier di-*cap* (bukan dihapus) agar tidak kehilangan data.

In [17]:
```
outlier_cols = ['annual_inc', 'dti', 'revol_bal', 'revol_util', 'delinq_2yrs']
available_outlier = [c for c in outlier_cols if c in df_clean.columns]

print("Penanganan Outlier (IQR capping):\n")
for col in available_outlier:
    Q1 = df_clean[col].quantile(0.25)
    Q3 = df_clean[col].quantile(0.75)
    IQR = Q3 - Q1
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR
    outliers_before = ((df_clean[col] < lower) | (df_clean[col] > upper)).sum()
    df_clean[col] = df_clean[col].clip(lower=lower, upper=upper)
    print(f"  {col}: {outliers_before:,} outlier di-cap (range: {lower:.2f} - {uppe

print(f"\nOutlier handling selesai. Shape: {df_clean.shape}")
```

```
Penanganan Outlier (IQR capping):

  annual_inc: 19,899 outlier di-cap (range: -20929.50 - 154882.50)
  dti: 43 outlier di-cap (range: -5.77 - 39.91)
  revol_bal: 22,879 outlier di-cap (range: -14467.00 - 41213.00)
  revol_util: 19 outlier di-cap (range: -14.05 - 127.95)
  delinq_2yrs: 83,302 outlier di-cap (range: 0.00 - 0.00)

Outlier handling selesai. Shape: (466285, 31)
```

5.5 Split Data & Feature Scaling

Membagi data menjadi training set (80%) dan testing set (20%), lalu melakukan StandardScaler pada fitur numerik. Stratified split digunakan agar proporsi label target tetap seimbang di kedua set.

In [18]:
```
X = df_clean.drop(columns=['loan_label'])
y = df_clean['loan_label']

print(f"Shape X: {X.shape}")
print(f"Shape y: {y.shape}")
print(f"Distribusi target:\n{y.value_counts()}")

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
                                                    random_state=42, stratify=y)

print(f"\nTraining set: {X_train.shape[0]:,} samples")
print(f"Testing set:  {X_test.shape[0]:,} samples")
```

```
Shape X: (466285, 30)
Shape y: (466285,)
Distribusi target:
loan_label
1    410953
0     55332
Name: count, dtype: int64

Training set: 373,028 samples
Testing set:  93,257 samples
```

In [19]:
```
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Pastikan tidak ada inf atau NaN setelah scaling
X_train_scaled = np.nan_to_num(X_train_scaled, nan=0.0, posinf=0.0, neginf=0.0)
X_test_scaled = np.nan_to_num(X_test_scaled, nan=0.0, posinf=0.0, neginf=0.0)

print("Feature scaling selesai (StandardScaler)")
print(f"   Mean training (sample): {X_train_scaled[:, 0].mean():.6f}")
print(f"   Std training (sample):  {X_train_scaled[:, 0].std():.6f}")
print(f"   Ada NaN di train: {np.isnan(X_train_scaled).any()}")
print(f"   Ada Inf di train: {np.isinf(X_train_scaled).any()}")
```

```
Feature scaling selesai (StandardScaler)
   Mean training (sample): -0.000000
   Std training (sample):  1.000000
   Ada NaN di train: False
   Ada Inf di train: False
```

6. Data Modelling

6.1 Model 1: Logistic Regression

Logistic Regression adalah algoritma klasifikasi wajib yang digunakan dalam project ini. Model ini cocok untuk klasifikasi biner dan memberikan probabilitas prediksi.

Parameter:

- `max_iter=1000` : Jumlah iterasi maksimum untuk konvergensi
- `solver='lbfgs'` : Optimizer yang efisien untuk dataset besar
- `class_weight='balanced'` : Menangani ketidakseimbangan kelas dengan memberikan bobot lebih pada kelas minoritas (BAD loan)
- `random_state=42` : Untuk reproduksibilitas

In [20]:
```
print("Melatih Logistic Regression...")
lr_model = LogisticRegression(
    max_iter=1000,
    random_state=42,
    solver='lbfgs',
    class_weight='balanced',
    n_jobs=-1
)
lr_model.fit(X_train_scaled, y_train)

y_pred_lr = lr_model.predict(X_test_scaled)
y_prob_lr = lr_model.predict_proba(X_test_scaled)[:, 1]

print("Logistic Regression selesai dilatih!")
print(f"\nClassification Report:\n")
print(classification_report(y_test, y_pred_lr, target_names=['BAD (0)', 'GOOD (1)']
```

```
Melatih Logistic Regression...
Logistic Regression selesai dilatih!

Classification Report:

              precision    recall  f1-score   support

     BAD (0)       0.19      0.63      0.29     11066
    GOOD (1)       0.93      0.63      0.75     82191

    accuracy                           0.63     93257
   macro avg       0.56      0.63      0.52     93257
weighted avg       0.84      0.63      0.70     93257
```

6.2 Model 2: Random Forest Classifier

Random Forest dipilih sebagai algoritma kedua karena kemampuannya menangani data yang kompleks, non-linear, dan memberikan feature importance.

Parameter:

- `n_estimators=200` : Jumlah decision tree dalam ensemble
- `max_depth=20` : Kedalaman maksimum tree untuk mencegah overfitting
- `min_samples_split=10` : Jumlah minimum sampel untuk melakukan split
- `class_weight='balanced'` : Menangani imbalanced class
- `random_state=42` : Untuk reproduksibilitas

In [21]:
```
print("Melatih Random Forest Classifier...")
rf_model = RandomForestClassifier(
    n_estimators=200,
    max_depth=20,
    min_samples_split=10,
    class_weight='balanced',
    random_state=42,
    n_jobs=-1
)
rf_model.fit(X_train_scaled, y_train)

y_pred_rf = rf_model.predict(X_test_scaled)
y_prob_rf = rf_model.predict_proba(X_test_scaled)[:, 1]

print("Random Forest selesai dilatih!")
print(f"\nClassification Report:\n")
print(classification_report(y_test, y_pred_rf, target_names=['BAD (0)', 'GOOD (1)']
```

```
Melatih Random Forest Classifier...
Random Forest selesai dilatih!

Classification Report:

              precision    recall  f1-score   support

     BAD (0)       0.26      0.19      0.22     11066
    GOOD (1)       0.90      0.93      0.91     82191

    accuracy                           0.84     93257
   macro avg       0.58      0.56      0.57     93257
weighted avg       0.82      0.84      0.83     93257
```

7. Evaluation

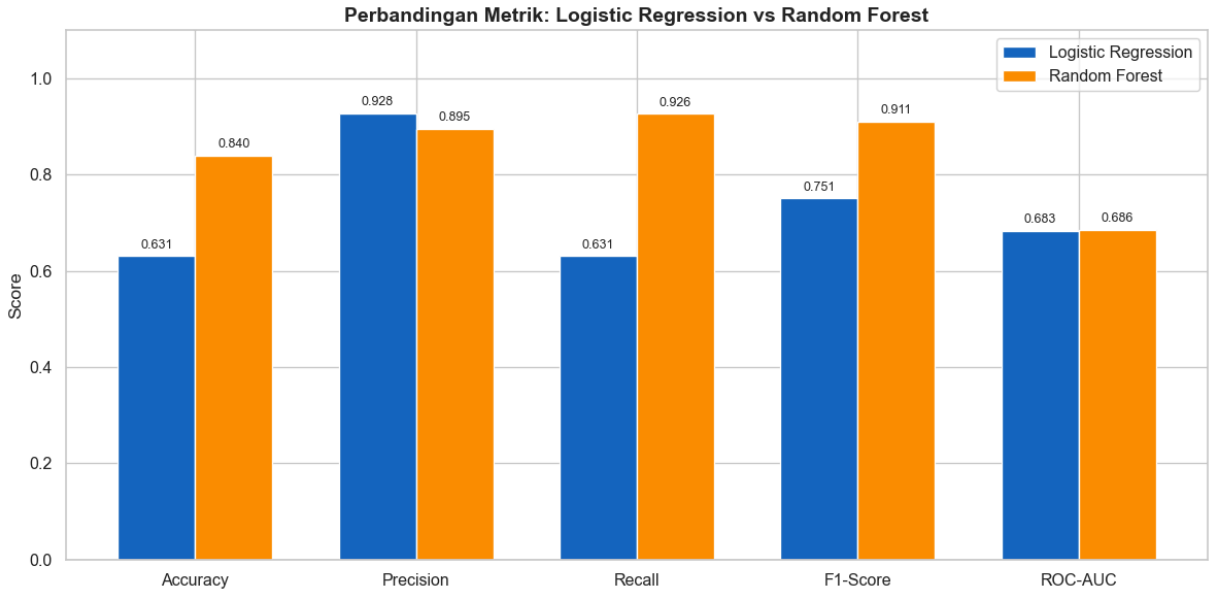
7.1 Perbandingan Metrik Evaluasi

Membandingkan kinerja kedua model menggunakan berbagai metrik evaluasi:

- **Accuracy**: Proporsi prediksi yang benar
- **Precision**: Ketepatan prediksi positif (GOOD loan)
- **Recall**: Kemampuan mendeteksi semua data positif
- **F1-Score**: Harmonic mean dari Precision dan Recall
- **ROC-AUC**: Area Under the ROC Curve, mengukur kemampuan diskriminasi model

In [22]:
```
metrics = {}
for name, y_pred, y_prob in [('Logistic Regression', y_pred_lr, y_prob_lr),
                              ('Random Forest', y_pred_rf, y_prob_rf)]:
    metrics[name] = {
        'Accuracy': accuracy_score(y_test, y_pred),
        'Precision': precision_score(y_test, y_pred, pos_label=1),
        'Recall': recall_score(y_test, y_pred, pos_label=1),
        'F1-Score': f1_score(y_test, y_pred, pos_label=1),
        'ROC-AUC': roc_auc_score(y_test, y_prob)
    }

metrics_df = pd.DataFrame(metrics).T
print("Perbandingan Metrik Evaluasi:\n")
print(metrics_df.round(4).to_string())

fig, ax = plt.subplots(figsize=(12, 6))
x = np.arange(len(metrics_df.columns))
width = 0.35
bars1 = ax.bar(x - width/2, metrics_df.iloc[0].values, width, label='Logistic Regre
               color='#1565C0', edgecolor='white')
bars2 = ax.bar(x + width/2, metrics_df.iloc[1].values, width, label='Random Forest'
               color='#FB8C00', edgecolor='white')
ax.set_ylabel('Score')
ax.set_title('Perbandingan Metrik: Logistic Regression vs Random Forest', fontsize=
ax.set_xticks(x)
ax.set_xticklabels(metrics_df.columns)
ax.legend()
ax.set_ylim(0, 1.1)
for b in [bars1, bars2]:
    for bar in b:
        ax.text(bar.get_x() + bar.get_width()/2., bar.get_height() + 0.01,
                f'{bar.get_height():.3f}', ha='center', va='bottom', fontsize=9)
plt.tight_layout()
plt.show()
```

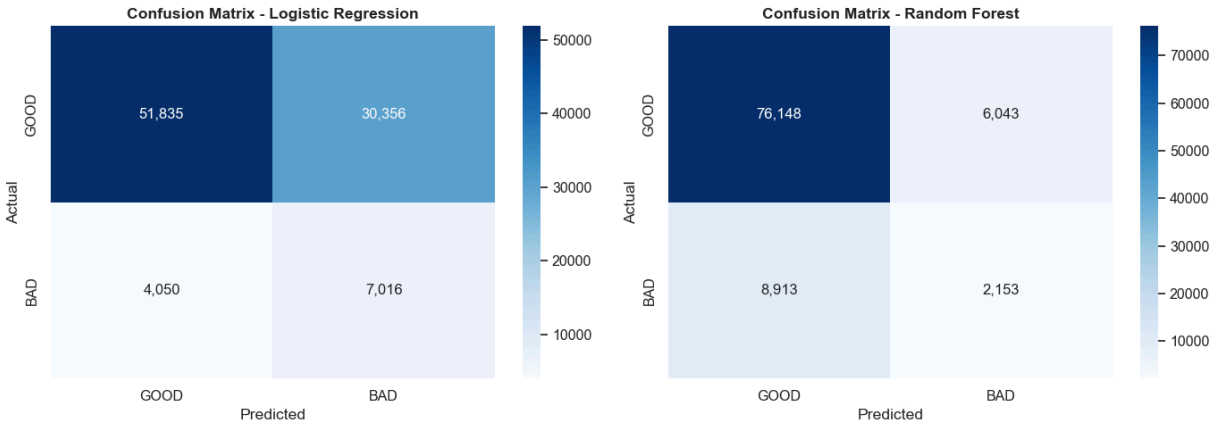
```
Perbandingan Metrik Evaluasi:

                     Accuracy  Precision  Recall  F1-Score  ROC-AUC
Logistic Regression    0.6311     0.9275  0.6307    0.7508   0.6834
Random Forest          0.8396     0.8952  0.9265    0.9106   0.6855
```



7.2 Confusion Matrix

Visualisasi confusion matrix untuk kedua model. Confusion matrix menunjukkan jumlah True Positive, True Negative, False Positive, dan False Negative.

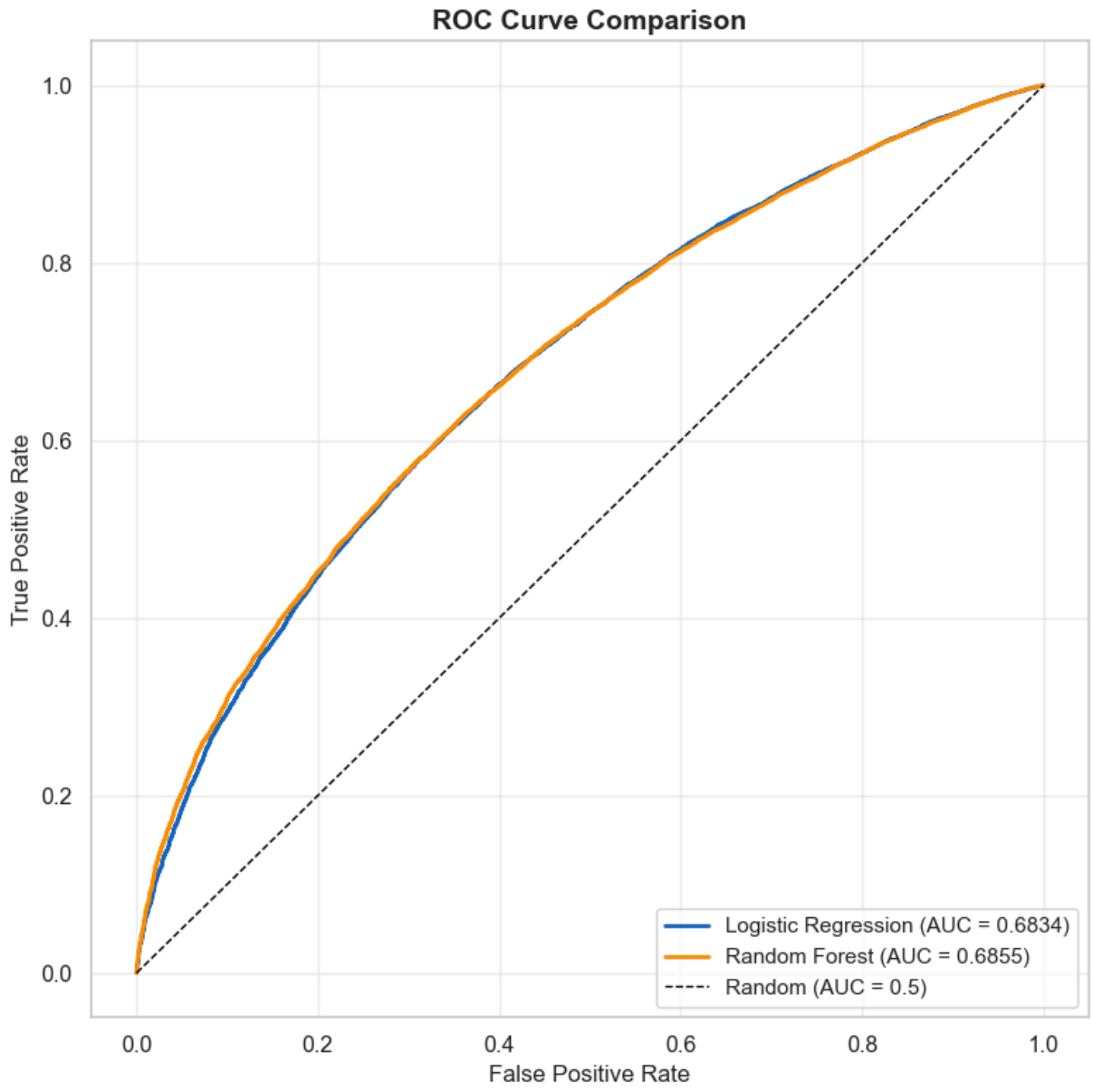
In [23]:
```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

for i, (name, y_pred) in enumerate([('Logistic Regression', y_pred_lr),
                                    ('Random Forest', y_pred_rf)]):
    cm = confusion_matrix(y_test, y_pred, labels=[1, 0])
    sns.heatmap(cm, annot=True, fmt=',', cmap='Blues', ax=axes[i],
                xticklabels=['GOOD', 'BAD'], yticklabels=['GOOD', 'BAD'])
    axes[i].set_title(f'Confusion Matrix - {name}', fontweight='bold')
    axes[i].set_xlabel('Predicted')
    axes[i].set_ylabel('Actual')

plt.tight_layout()
plt.show()
```



7.3 ROC Curve

Membandingkan kurva ROC (Receiver Operating Characteristic) kedua model. Semakin besar area di bawah kurva (AUC), semakin baik kemampuan model dalam membedakan kelas positif dan negatif.

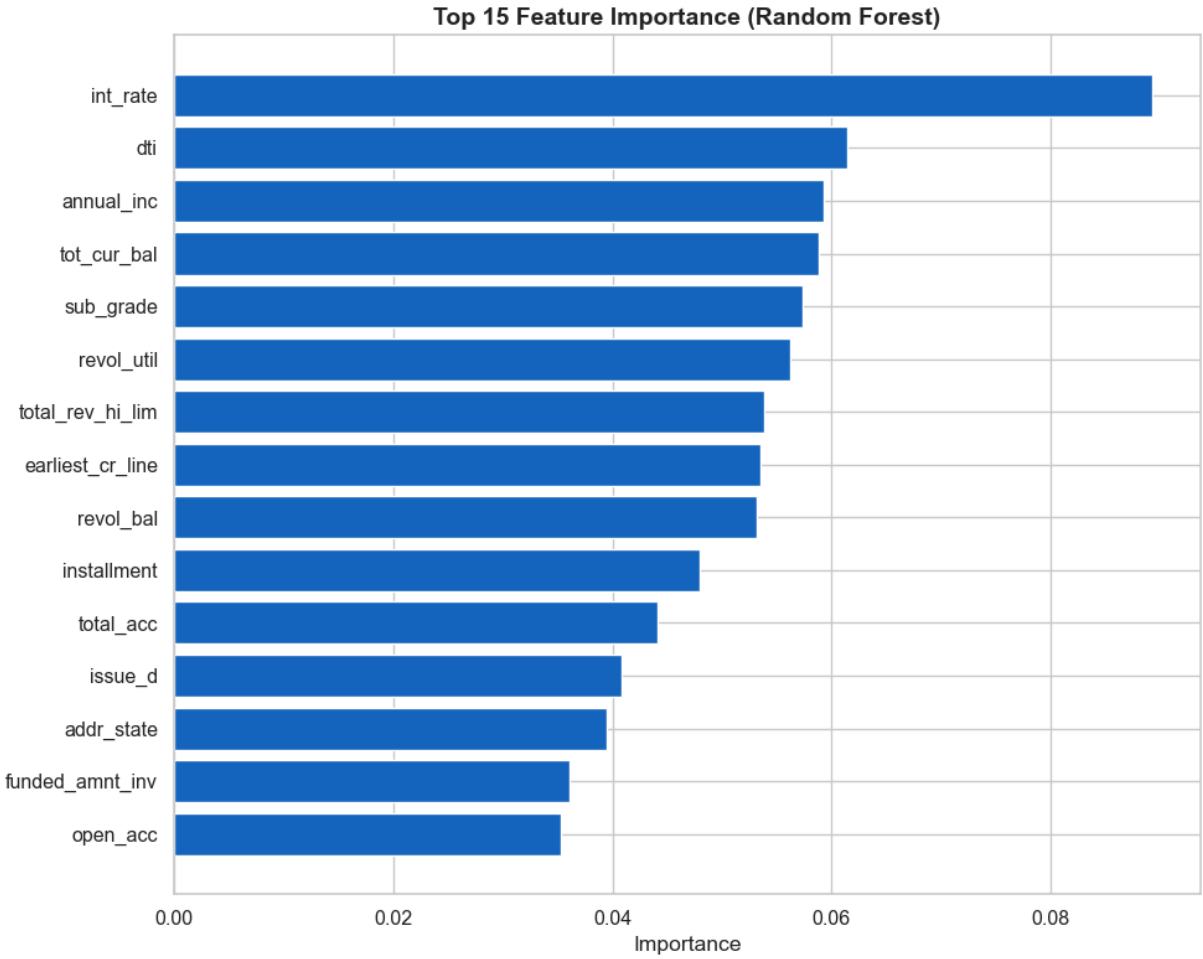
In [24]:
```
fig, ax = plt.subplots(figsize=(8, 8))

for name, y_prob, color in [('Logistic Regression', y_prob_lr, '#1565C0'),
                            ('Random Forest', y_prob_rf, '#FB8C00')]:
    fpr, tpr, _ = roc_curve(y_test, y_prob)
    auc = roc_auc_score(y_test, y_prob)
    ax.plot(fpr, tpr, color=color, lw=2, label=f'{name} (AUC = {auc:.4f})')

ax.plot([0, 1], [0, 1], 'k--', lw=1, label='Random (AUC = 0.5)')
ax.set_xlabel('False Positive Rate', fontsize=12)
ax.set_ylabel('True Positive Rate', fontsize=12)
ax.set_title('ROC Curve Comparison', fontsize=14, fontweight='bold')
ax.legend(loc='lower right', fontsize=11)
ax.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



7.4 Feature Importance (Random Forest)

Melihat fitur-fitur yang paling berpengaruh dalam prediksi model Random Forest. Feature importance membantu mengidentifikasi variabel mana yang paling penting dalam menentukan good vs bad loan.

In [25]:
```
feat_imp = pd.DataFrame({
    'Feature': X.columns,
    'Importance': rf_model.feature_importances_
}).sort_values('Importance', ascending=True).tail(15)

fig, ax = plt.subplots(figsize=(10, 8))
ax.barh(feat_imp['Feature'], feat_imp['Importance'], color='#1565C0', edgecolor='wh
ax.set_xlabel('Importance')
ax.set_title('Top 15 Feature Importance (Random Forest)', fontsize=14, fontweight='
plt.tight_layout()
plt.show()
```



7.5 Cross Validation

Melakukan 5-Fold Cross Validation untuk memvalidasi stabilitas dan konsistensi model. Cross validation membagi data training menjadi 5 bagian (fold), melatih model pada 4 fold dan menguji pada 1 fold secara bergantian.

In [26]:
```
print("Melakukan 5-Fold Cross Validation...\n")

for name, model in [('Logistic Regression', lr_model), ('Random Forest', rf_model)]
    cv_scores = cross_val_score(model, X_train_scaled, y_train, cv=5, scoring='accu
    print(f"{name}:")
    print(f"   CV Scores: {cv_scores.round(4)}")
    print(f"   Mean CV Accuracy: {cv_scores.mean():.4f} (+/- {cv_scores.std()*2:.4f
```

```
Melakukan 5-Fold Cross Validation...

Logistic Regression:
   CV Scores: [0.6283 0.631  0.6304 0.6278 0.6287]
   Mean CV Accuracy: 0.6293 (+/- 0.0025)

Random Forest:
   CV Scores: [0.8431 0.8442 0.8449 0.8428 0.8426]
   Mean CV Accuracy: 0.8435 (+/- 0.0017)
```

8. Kesimpulan

Ringkasan Hasil:

1. **Data Understanding**: Dataset terdiri dari 466,285 data pinjaman dengan 75 kolom. Kolom `loan_status` digunakan sebagai dasar pembuatan label target (GOOD vs BAD loan).

2. **EDA**:

   - Distribusi data menunjukkan ketidakseimbangan antara GOOD dan BAD loan (~88% vs ~12%)
   - Variabel seperti `int_rate`, `grade`, dan `dti` menunjukkan perbedaan signifikan antara good dan bad loan
   - Korelasi antar variabel teridentifikasi (misalnya `loan_amnt` dan `installment` berkorelasi tinggi)
3. **Data Preparation**:

   - Missing values ditangani dengan median (numerik) dan modus (kategorikal)
   - Kolom dengan >50% missing values dihapus
   - Outlier ditangani dengan metode IQR capping
   - Variabel kategorikal di-encode menggunakan Label Encoding
   - Data dibagi 80:20 untuk training dan testing dengan stratified sampling
   - Feature scaling menggunakan StandardScaler
4. **Modelling & Evaluation**:

   - Dua model digunakan: **Logistic Regression** (wajib) dan **Random Forest**
   - Parameter `class_weight='balanced'` digunakan untuk menangani imbalanced data
   - Evaluasi dilakukan menggunakan Accuracy, Precision, Recall, F1-Score, dan ROC-AUC
   - Cross-validation 5-fold dilakukan untuk validasi stabilitas model

Rekomendasi:

- Model dengan performa terbaik dapat dipilih berdasarkan kebutuhan bisnis
- Jika fokus pada minimalisir false negative (mendeteksi semua bad loan), prioritaskan **Recall**
- Jika fokus pada keseimbangan, gunakan **F1-Score** atau **ROC-AUC** sebagai metrik utama
- Untuk peningkatan lebih lanjut, bisa dicoba: SMOTE oversampling, feature engineering lebih lanjut, atau hyperparameter tuning dengan GridSearchCV